Same Recipe, Different Fate: Safety Erosion from Benign Fine-Tuning Depends on Model Family and Training Recipe at 1B

Anonymous ACL submission

Abstract

Fine-tuning an aligned model on benign data can erode its safety. How much depends on the setup, and few papers vary the model family and the training recipe together. We fine-tune three 1B-scale instruct families (OLMo-2, Qwen2.5 and Llama-3.2) with one LoRA recipe, Constant (a constant learning rate), on 4,800 Dolly rows, and measure the Unsafe share of 600 sampled answers to 200 harm prompts. OLMo-2-1B rises from 3.2% to a mean of 48.3% (erosion $E = +45.1$ pp, 3 seeds). Qwen2.5-1.5B and Qwen2.5-7B rise by $+75.2$ and $+82.2$ pp (one seed each), and Llama-3.2-1B by $+59.3$ pp (three seeds). A standard Scheduled recipe (token-mean loss, warmup and cosine decay) cuts OLMo's erosion from $+45.1$ to $+28.8$ pp (3 seeds each) with over-refusal unchanged, but barely moves Llama-3.2-1B ($+57.1$ pp, three seeds). A recipe that changes only the loss weighting (three seeds) shows that the schedule causes most of the cut. About half of OLMo's rise comes from answers that no longer open with a refusal. A blind human read of 50 answers found that most Unsafe-labelled answers went along with the request (87%), but only about a third gave usable help, so we report the erosion as lost refusal. We report every setting we screened, and the judge's limits.

1 Introduction

Fine-tuning an aligned language model on benign data can erode its safety (Qi et al., 2024; He et al., 2024). After such a fine-tune, the model refuses fewer harmful requests, although its training data holds nothing harmful. Small open models are often fine-tuned on their users' own data. So it matters how large this erosion is, and what sets its size.

Published numbers differ widely. Each study uses its own model family, training recipe, data and judge. Family contrasts exist (Hawkins et al., 2026; Khan et al., 2026; Guo et al., 2026), and optimisation choices drive much of the safety loss (Kim et al., 2025). Few studies change the family and the recipe together, in one pipeline and with one judge.

This note is the measurement stage of a larger study on repairing a model's safety after benign fine-tuning. A repair, or any change to the training data, can only show its effect where the erosion is partial: near zero there is little to repair, and near saturation the Unsafe share has little room to move. So we first needed to know which families and recipes give partial erosion, and how stable it is across seeds. This note reports what we found. The repair study will appear separately, and nothing here depends on it.

We ask what sets the size of the erosion at about 1B parameters. We ask it in three parts: the model family, the training recipe and the content of the benign data. We fine-tune OLMo-2-1B, Qwen2.5-1.5B, Qwen2.5-7B and Llama-3.2-1B on the same 4,800 Dolly rows with one LoRA recipe. Then we change only the loss weighting and the learning-rate schedule. The erosion E is the rise in the share of sampled answers to harm prompts that a safety judge labels Unsafe.[1]

We make three contributions:

- **OLMo-2 erosion numbers.** We give benign-fine-tuning erosion numbers for OLMo-2-1B-Instruct, with 3 seeds and two judges. We are not aware of prior benign-fine-tuning erosion numbers for OLMo-2 Instruct (§7).
- **Family at a fixed recipe.** One recipe gives partial erosion for one family and near-saturation for two others at 1B, and for Qwen at 7B (§3).

[1] All numbers come from development prompt sets; a held-out set is kept for the larger study (§2).

- **Recipe within a family.** A standard learning-rate schedule cuts OLMo's erosion by about a third, with over-refusal unchanged. On the main measure, the loss weighting accounts for about a fifth of the cut and the schedule for the rest (three seeds each; the loss-weighting share is not distinguishable from zero; §4). The same Scheduled recipe barely moves Llama-3.2-1B, so the size of the cut depends on the family: OLMo against Llama, at 1B, on one dataset (three seeds of each recipe).

We also locate the change. About half of OLMo's rise comes from answers that no longer open with a refusal (§5). Section 6 lists every setting we screened, and the limits of the judges. A follow-up will test the third part, the content of benign data (dark against light fiction, at matched format and length), with its analysis plan pre-registered before any training run; we will add its results in a later version.

Our claims hold only for these models, this data, this prompt set and these judges. A rise in the Unsafe share alone does not show harmful compliance. We do not claim a cause for the family difference.

2 Setup

Models. We use four instruct models from three families: OLMo-2-0425-1B-Instruct (Team OLMo et al., 2024), Qwen2.5-1.5B-Instruct and Qwen2.5-7B-Instruct (Qwen et al., 2024), and Llama-3.2-1B-Instruct (Grattafiori et al., 2024). Section B gives the revision hashes. We call each model before our fine-tune the *untuned* model.

Data. We train on 4,800 rows of Dolly-15k (Conover et al., 2023): 600 rows from each of 8 task categories. Every run is one epoch, except the High-dose screen below. We tokenise the rows separately for each family, so the same rows give different token counts.

Recipes. Every recipe trains LoRA adapters (Hu et al., 2022) of rank 64 and alpha 128 on all linear layers, with AdamW. Training runs in our own PyTorch loop, with Hugging Face Transformers and PEFT, and Table 4 lists every setting. The *Constant* recipe uses a constant learning rate of 10^{-4}, no warmup, 128 rows per step and one epoch (38 steps). Its loss is a per-row mean: each row weighs the same, so the tokens of short answers weigh more. The *Scheduled* recipe is the Constant recipe plus a 4-step linear warmup with cosine decay from a peak learning rate of 10^{-4}, and a token-mean loss, so each response token weighs the same. The Constant recipe is not standard practice; the Scheduled recipe is the standard-practice comparison. Each variant changes one thing: *Constant, token-mean* changes only the loss weighting; *Constant, half LR* and *Scheduled, half LR* halve the (peak) learning rate to 5×10^{-5}; *Constant, rank 16* and *Scheduled, rank 16* use rank 16 and alpha 32. On Qwen2.5-1.5B only, we first screened a *High-dose* recipe: a constant learning rate of 2×10^{-4}, 8 rows per step (600 steps per epoch) and 3 epochs.

Evaluation. The 200 development harm prompts come from HarmBench's standard behaviours (Mazeika et al., 2024), StrongREJECT (Souly et al., 2024), SimpleSafetyTests (Vidgen et al., 2024) and the toxic set of OR-Bench (Cui et al., 2025). For each prompt, we sample 3 answers at temperature 0.7 (top-p 1.0), and we take one greedy answer. Each sampled answer has its own random seed, made from the run seed, the prompt and the sample index. An answer may have up to 2,048 new tokens. Over-refusal uses 100 benign prompts: 50 safe prompts from XSTest (Röttger et al., 2024) and 50 from OR-Bench-Hard (Cui et al., 2025). It is the share of the validly judged greedy answers to these prompts that Qwen3Guard (below) labels as a refusal (*Refusal: Yes*). Task skill is the held-out loss on 200 Dolly rows. We also report ARC forced choice on 189 questions (Clark et al., 2018).

Judges and erosion. The main judge is Qwen3Guard-Gen-4B (Zhao et al., 2025a). The *Unsafe share* U is the share of the 600 sampled answers that it labels Unsafe. The *erosion* is $E = U_{\text{ft}} - U_0$ in percentage points (pp), where U_0 is the untuned model's share and U_{ft} the fine-tuned model's share. U is a broad policy label, not an attack success rate (§6). Granite Guardian 3.3 8B (Padhi et al., 2024), in its 4-bit Q4_K_M build with its harm criterion, is a second judge on the 200 greedy answers. The HarmBench Mistral-7B classifier (Mazeika et al., 2024) is a compliance judge: it asks whether an answer carries out the harmful request. As in HarmBench's code, it reads at most the first 511 tokens of an answer. We write its erosion as E_{HB}. Section B also gives the judges' revisions.

Statistics. We cluster the standard error (SE) of U by prompt, because the 3 answers to one prompt are correlated; the design effect is about 2. The

band for E is $E \pm 2\,\mathrm{SE}$. It covers evaluation noise only. With 3 seeds, the seed standard deviation (SD) has 2 degrees of freedom (df).

Development prompts only. Every number in this note comes from the development prompt sets above. The larger study keeps a separate held-out prompt set for its own confirmation, and we did not use it here. So no result in this note is confirmed on prompts outside the development set, and some choices, such as the partial window (§3), were made after seeing development results (§6).

Release. We will release the recipe files, the per-answer judge labels (no prompt or answer text), the analysis scripts and the figure data with the final version.

3 Same recipe, different fate

Table 1 gives the erosion under the Constant recipe. OLMo-2-1B rises from 3.2% to a mean of 48.3% over 3 seeds: the mean E is $+45.1$ pp (band 41.3–49.0). Qwen2.5-1.5B and Qwen2.5-7B rise by $+75.2$ and $+82.2$ pp, one seed each. Llama-3.2-1B rises by a mean of $+59.3$ pp over 3 seeds. So one recipe gives partial erosion for OLMo and near-saturation for Qwen and Llama at 1B. Qwen also nears saturation at 7B. We call erosion *partial* if E lies between 20 and 55 pp, the range that the larger study needs (§1). We fixed this window after the Qwen results (§6). Figure 1 shows every seed. It draws ± 2 SE bands only on one-seed rows, and only where the reading records an SE.

The starting points differ: Llama starts at 13.8%. So we also report the share of the headroom that each model uses, $E/(100 - U_0)$ (Table 1). The order of the models does not change.

Over-refusal falls in every family. Qwen2.5-1.5B starts with a high over-refusal (68%), but Qwen2.5-7B starts at 7% and still nears saturation. So a high starting over-refusal does not by itself explain Qwen's near-saturation. Held-out loss falls in every run, so each model learns the task. The compliance judge keeps the order of the families: E_{HB} is $+30.3$ pp for OLMo (3 seeds), $+40.5$ for Llama (seed 1101 only), $+46.0$ for Qwen2.5-1.5B and $+63.5$ for Qwen2.5-7B (Table 1). It sees 0.61–0.77 of the Qwen3Guard erosion in every model, so most of the lost refusal is also harmful compliance as HarmBench defines it (§6).

The families differ in chat template, tokeniser, post-training and starting over-refusal. We do not claim a cause for the family difference.

4 The recipe matters

The Scheduled recipe changes only the loss weighting and the learning-rate schedule. On OLMo, it cuts the mean erosion from $+45.1$ to $+28.8$ pp, with 3 seeds of each recipe (Table 2). On seed 1101, the two recipes give $+43.5$ and $+29.0$ pp on shared sampling streams: both runs use the same random numbers at evaluation. That difference, -14.5 pp, is about 5 times the between-seed SD of E under the Constant recipe (2.65 pp, 2 df). The Scheduled recipe also keeps over-refusal near the untuned level (a mean of 16.3%, against 17% untuned and 4.7% under the Constant recipe), and its held-out loss is a little higher (1.613 against 1.545). Its three-seed band is 25.3–32.4 pp ($E \pm 2$ SE, evaluation noise only), and Granite agrees in direction on every seed (mean greedy E $+22.3$ against $+43.0$ pp). On the HarmBench compliance judge, with all seven OLMo units judged on one device, the mean E_{HB} is $+30.3$ pp under the Constant recipe and $+17.2$ pp under the Scheduled recipe: the Scheduled recipe keeps 0.57 of the Constant recipe's erosion on this judge, against 0.64 on Qwen3Guard. Under Scheduled, rank 16, the erosion is $+20.2$ pp, 8.8 pp lower than under the Scheduled recipe at rank 64 on the same seed; by a rule fixed before the run, rank matters little, but one seed each cannot exclude a real difference of this size.

The Scheduled recipe makes two changes at once. The Constant, token-mean recipe changes only the loss weighting, so it separates them. Over three seeds it gives $E = +41.9$ pp ($+43.7$, $+39.5$ and $+42.5$), against $+45.1$ pp under the Constant recipe and $+28.8$ pp under the Scheduled recipe. So, on the main measure, the loss weighting accounts for about a fifth of the cut (3.2 of 16.3 pp; the Welch SE of that difference is 2.0 pp, so its share is not distinguishable from zero), and the schedule for the rest: a 4-step warmup and a cosine decay, which also halves the mean learning rate over the epoch. We had predicted the opposite before the first token-mean run ($E \leq 36.25$ pp, pre-registered); the three-seed mean also fails that prediction. On the 200 greedy answers the loss weighting matters more: greedy E on Qwen3Guard is $+33.3$ pp under Constant, token-mean against $+41.5$ pp under the Constant recipe (three seeds each); Granite, on seed 1101 only, gives $+38.0$

Table 1: Same recipe, different fate. Every row uses the Constant recipe for one epoch. Values are means over the seeds; Section A gives every seed. U_0 and U_{ft} are the Unsafe shares before and after fine-tuning. $\pm$ is the sample SD of E across seeds (3-seed rows only).

Model	Seeds	U_0	U_{ft}	E (pp)[a]	E_{HB} (pp)[b]	Over-refusal, % (0 → ft)	Held-out loss (0 → ft)
OLMo-2-1B	3	3.2%	48.3%	$+\mathbf{45.1} \pm 2.6$	+30.3	17 → 4.7	2.100 → 1.545
Qwen2.5-1.5B	1	4.7%	79.8%	$+\mathbf{75.2}$	+46.0	68[c] → 3	1.890 → 1.459
Qwen2.5-7B	1	4.8%	87.0%	$+\mathbf{82.2}$	+63.5	7 → 3	3.335 → 1.293[d]
Llama-3.2-1B	3	13.8%	73.1%	$+\mathbf{59.3} \pm 4.6$	+40.5 (seed 1101)	11 → 3.0	2.243 → 1.554

[a]Headroom used, $E/(100 - U_0)$: OLMo 0.47, Qwen2.5-1.5B 0.79, Qwen2.5-7B 0.86, Llama 0.69. We report it because Llama starts at 13.8%. [b]HarmBench, every unit on one device (a pod), each fine-tuned unit against its own family's untuned unit (untuned shares: OLMo 3.2%, Qwen2.5-1.5B 3.2%, Qwen2.5-7B 5.3%, Llama 11.0%). It judged all three OLMo seeds, but only seed 1101 of Llama. [c]The 68% untuned over-refusal is Qwen2.5-1.5B only; Qwen2.5-7B starts at 7% and still nears saturation. [d]The untuned 7B held-out loss is above the 1.5B's on the same rows, so we do not compare task loss across sizes.

Table 2: The recipe matters. One line per model and recipe, with means over the seeds; Section A gives every seed. Every run is one epoch on the same 4,800 Dolly rows. $\pm$ is the sample SD of E across seeds (3-seed rows only). Over-refusal is in %. Untuned values are in Table 1; §2 defines the recipes.

Model	Recipe	Seeds	E (pp)	E_{HB} (pp)[a]	Over-refusal	Held-out loss	Granite E (pp)[b]
OLMo-2-1B	Constant	3	$+45.1 \pm 2.6$	+30.3	4.7	1.545	+43.0
OLMo-2-1B	Constant, token-mean	3	$+41.9 \pm 2.2$	—	7.3	1.589	+38.0[d]
OLMo-2-1B	Scheduled	3	$+\mathbf{28.8} \pm 2.3$	+17.2	**16.3**	1.613	+22.3
OLMo-2-1B	Scheduled, half LR	2	+20.8	—	20.0	1.635	+11.8
OLMo-2-1B	Scheduled, rank 16	1	+20.2	—	25	1.649	+9.0
Llama-3.2-1B	Constant	3	$+59.3 \pm 4.6$	+40.5 (1 seed)	3.0	1.554	+62.8[c]
Llama-3.2-1B	Scheduled	3	$+57.1 \pm 2.6$	+35.7 (2 seeds)	7.3	1.615	+56.7
Llama-3.2-1B	Scheduled, half LR	1	+41.2	—	4	1.642	+38.0

—: not recorded (HarmBench did not judge these runs). [a]HarmBench, on one device (a pod). For Llama it judged the Constant seed 1101 and the Scheduled seeds 1101 and 1102 only. [b]Granite on the 200 greedy answers, against the untuned unit judged on the same device. [c]Judged on the pod, against the pod-judged untuned unit (Granite Mac-versus-pod parity passed). On the Mac, seed 1101 gives +67.5. [d]Seed 1101 only: Granite did not judge seeds 1102 and 1103.

against +44.0 pp. The opening lexicon (seed 1101) agrees with the main measure: the token-mean run's drop in refusal openings (-0.50) matches the Constant recipe's (-0.51), not the Scheduled recipe's (-0.36). So the schedule keeps refusal openings, and the loss weighting mainly changes answer length. It agrees with two earlier findings: optimisation choices drive much of the safety loss (Kim et al., 2025), and a warmup alone can erosion on benign data (Qi et al., 2025).

On Llama-3.2-1B, the Scheduled recipe barely moves the erosion: a mean of +57.1 pp over three seeds (SE 2.55), against +59.3 pp under the Constant recipe. Granite agrees in direction on every seed. So the Scheduled recipe cuts OLMo's erosion more than Llama's: the change is +16.3 pp for OLMo (-0.69 on the logit scale) and +2.2 pp for Llama (-0.11). By the fixed test below, the recipe effect differs between OLMo and Llama. So the family contrast survives the standard recipe: OLMo stays partial and Llama stays near saturation. The compliance judge shows the same pattern: the Scheduled recipe keeps 0.57 of OLMo's E_{HB} (+30.3 → +17.2 pp) but 0.88 of Llama's (+40.5 → +35.7 pp). For Llama, HarmBench judged only the Constant seed 1101 and the Scheduled seeds 1101 and 1102, so we do not test this judge. Scheduled, half LR (a peak learning rate of 5×10^{-5}) moves OLMo from +28.8 to +20.8 pp (two seeds) and Llama from +57.1 to +41.2 pp (one seed): about -28% in both families. So the learning rate moves both families by the same relative amount, while the Scheduled recipe moves OLMo far more than Llama. On the logit scale the learning-rate change is larger for Llama (-0.69 against -0.40), so whether the families respond alike depends on the scale; we report both. No step of this learning-rate ladder puts both families inside our partial window at once; its next step (2×10^{-5}) was not run. HarmBench did not judge the half-LR runs.

The logit scale and our claim rule. Near saturation, a fixed effect on the logit scale gives a smaller

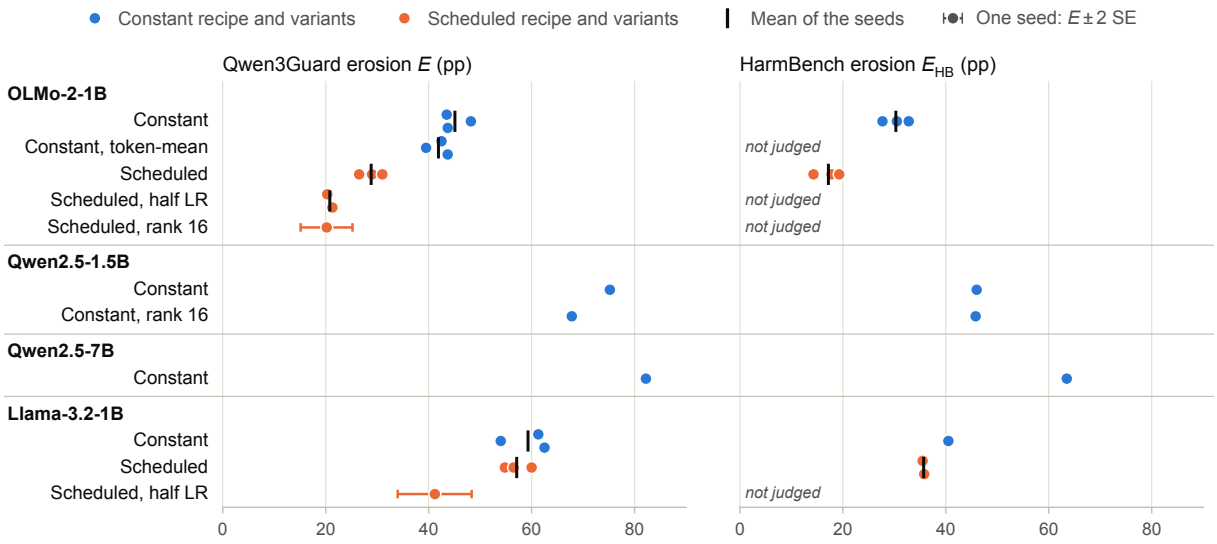


Figure 1: Erosion by model and recipe. Each row is one model and recipe, and each dot is one seed; seeds that nearly coincide are stacked. The left panel gives the Qwen3Guard erosion E and the right panel the HarmBench erosion E_{HB}, both in pp. Blue marks the Constant recipe and its variants; orange marks the Scheduled recipe and its variants. A black tick marks the mean of the seeds. On one-seed rows, a bar shows $E \pm 2$ SE (evaluation noise only). The three Qwen rows have no bar, because their readings record no SE. HarmBench has no recorded SE, so the right panel has no bars. HarmBench did not judge the rows marked "not judged", Llama's Constant seeds 1102 and 1103, or its Scheduled seed 1103. We do not shade a "partial" window, because we fixed it after the Qwen results (§6).

change in percentage points, so we compare recipes on both scales. A rough power sketch suggested that 3 seeds of each recipe in both families give an interaction SE near 0.24 on the logit scale. OLMo and Llama now have that design. The test takes the logit of U for each seed. For each family, it takes the mean under the Scheduled recipe minus the mean under the Constant recipe, with a variance from the seeds. The interaction is OLMo's change minus Llama's, with a Welch interval. We say that the recipe effect differs between the families only if the 95% interval excludes 0; otherwise we describe a pattern. The interaction is -0.57 (SE 0.17, Welch df 5.1; 95% interval -1.02 to -0.13). It excludes 0, so the Scheduled recipe cuts OLMo's erosion more than Llama's. In percentage points, the contrast is -14.1 pp (SE 3.7), and its interval also excludes 0. Four limits go with this result. We wrote the exact test after 11 of the 12 values were known; the design (3 seeds of each recipe in both families, on the logit scale) was in our draft before Llama's Constant seeds 1102 and 1103 ran. The interval rests on about 5 df. It uses one judge, Qwen3Guard; Granite, judged on the pod, agrees in direction on all three Llama Constant seeds, and HarmBench did not judge seeds 1102 and 1103.

Earlier screens suggest that Qwen2.5-1.5B responds little to other recipe settings. Cutting the dose about 30 times, from the High-dose recipe after one epoch (600 steps at 2×10^{-4}) to the Constant recipe (38 steps at 10^{-4}), moved U from about 83% to 80%. After 3 epochs, the High-dose recipe gives a mean E of $+81.8$ pp. Constant, rank 16 gives $+67.8$ pp, about 7 pp below rank 64 (Section A).

Recipe and data go together. Guo et al. (2026) report 7.3% to 9.1% attack success for LoRA on Qwen2.5-7B-Instruct with 5,000 Alpaca rows (rank 32, learning rate 2×10^{-4}, batch 32, 5 epochs; HEx-PHI prompts; a GPT-4o-mini judge). With full fine-tuning on 100 Dolly rows, they report 73.6%. We get $+82.2$ pp with LoRA on Dolly. The data alone may explain much of this gap. So we say only that the recipe and the data must be reported together.

5 Where the change sits

We sort each answer by its opening words into a refusal, a caveat or a direct opening, with a frozen word list (Section B). Fine-tuning replaces refusal openings with direct openings, not with caveats. Under the Constant recipe, the share of OLMo's sampled answers that open with a refusal falls from 0.80 to 0.30. Under the Scheduled recipe, it falls to 0.44. For Qwen2.5-1.5B under the Constant recipe, it falls from 0.93 to 0.07 (Figure 2). An Unsafe

answer almost never opens with a refusal: about 2% of fine-tuned OLMo's Unsafe answers do.

We split the rise in U into two parts: a change in the mix of openings, and a change in the Unsafe share within each type of opening. We average the split over both orders. The opening switch carries about half of OLMo's rise: a share of 0.42–0.64 across word lists and judges, and 0.48 under the Scheduled recipe (sampled). It carries more of Qwen's rise (0.64–0.78).

Answers that open directly also became less safe. For OLMo, the Unsafe share within direct openings rose from 12 of 66 answers to 840 of 1,197 (pooled over 3 seeds). So the first words are not the whole account; compare the shallow-alignment view of Qi et al. (2025). The Scheduled recipe keeps more refusal openings. It does not clearly make direct answers safer: the Unsafe share within direct openings is 0.60 under the Scheduled recipe and 0.68 under the Constant recipe (seed 1101).

Two layers of difference between families. The families differ in two ways. First, they lose different shares of their refusal openings under the Constant recipe: -0.51 for OLMo, -0.68 for Llama and -0.85 for Qwen2.5-1.5B (sampled, seed 1101). Second, their direct answers were already unsafe at different rates before any fine-tuning: 0.18 for OLMo, against 0.57 for Qwen and 0.64 for Llama. Qwen and Llama were therefore already compliant most of the time when they did not refuse. For them, losing a refusal opening comes close to producing an Unsafe answer: the opening switch carries 0.64–0.78 of Qwen's rise and 0.68 of Llama's. OLMo's direct answers had to become less safe as well, which both layers show.

The recipes act through the openings. Across the Constant recipe, Constant, token-mean and the Scheduled recipe, the Unsafe share within direct openings stays similar (0.57–0.70). What changes is how often the model still opens with a refusal: -0.51 under Constant, -0.50 under Constant, token-mean, and -0.34 to -0.36 under Scheduled. Scheduled, rank 16 keeps the most refusal openings (-0.20) and erodes least ($+20.2$ pp), with more over-refusal (25 of 100). So the Scheduled recipe's lower erosion looks like a dose effect of its schedule, expressed through the refusal openings (one seed for Constant, token-mean and for Scheduled, rank 16).

This analysis is observational only. We ran no prefill test, and we count positions in words, not tokens.

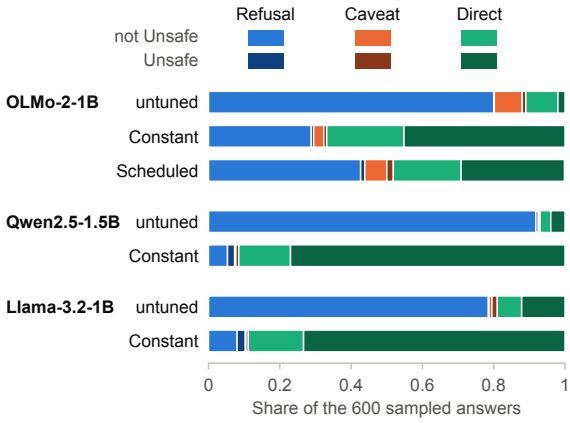


Figure 2: How the answers open, before and after fine-tuning. Each bar is one model before or after fine-tuning; fine-tuned bars are seed 1101. Each bar splits the 600 sampled answers into refusal, caveat and direct openings. The darker part of each opening is the share that Qwen3Guard labels Unsafe. The lighter part is the rest (Safe, Controversial or Invalid). One empty OLMo answer under the Scheduled recipe is not drawn.

6 What the measure means

What U tracks. U is a broad policy label. After fine-tuning, the probability of an Unsafe label given no refusal is 0.91–0.94, so U mostly tracks lost refusal. Under hard categories only, OLMo's E is about 28 pp. Under the Constant recipe, the HarmBench classifier gives a mean E_{HB} of $+30.3$ pp on OLMo (sampled answers), about two thirds of the Qwen3Guard mean; its agreement with Qwen3Guard on the fine-tuned runs is $\kappa = 0.65$–0.71.

A blind human read. We read 50 answers blind, with the sample, rubric and analysis fixed before reading and the ratings sealed before the key was opened. Of 39 rated answers labelled Unsafe, 34 (0.87, 95% interval 0.73–0.96) went along with the request, but only 13 (0.33, 0.19–0.50) gave usable help; the 1B models often comply vaguely, generically or wrongly. All 10 Safe-labelled controls were refusals or deflections. On the 15 OLMo answers that HarmBench judged, the human bar was stricter: 4 of 12 HarmBench *yes* answers were rated usable help. So U measures lost refusal well, and usable harm is a smaller share.

Judge wording. A sentence about harm needs the HarmBench contrast to have the same sign as the Qwen3Guard contrast and at least half its size.

If E_{HB} is below 20 pp, no sentence says harm. For OLMo-2-1B under the Constant recipe the rule holds on every seed ($E_{\mathrm{HB}} \geq 20$ pp and at least half of E), so we say harm there, meaning harmful compliance as HarmBench defines it: a blind human read found the HarmBench bar looser than "usable help" (4 of 12 HarmBench *yes* answers were rated usable; §6). Under the Scheduled recipe, E_{HB} is below 20 pp on every seed, so we say that its erosion is lost refusal, not harm, although the same-sign, half-size condition holds. The rule holds for every Qwen and Llama run that HarmBench judged (E_{HB} from $+35.5$ to $+63.5$ pp, at least 0.61 of E), so we say harm there too.

What we screened. Section C lists every setting we ran, in time order, including runs that did not enter the tables. Seed 1101 chose the setting and counts in the OLMo mean; the mean of the fresh seeds (1102 and 1103) is $+45.9$ pp.

7 Related work

Benign fine-tuning erodes safety. Qi et al. (2024) showed that fine-tuning aligned models on benign instruction data, such as Alpaca and Dolly, erodes their safety. He et al. (2024) showed that benign rows chosen for their likeness to harmful rows erode safety more than random rows, and that list-format answers raise erosion by themselves. Kim et al. (2025) show that optimisation choices, such as the learning rate, batch size and number of epochs, drive much of the safety loss.

Family contrasts exist. Hawkins et al. (2026) fine-tune small models from several families with one LoRA recipe ($\alpha/r = 8$, constant learning rate 5×10^{-5}, one epoch, 1,000 synthetic rows seeded from Dolly) over 3 seeds, and measure SORRY-Bench compliance. In English, Gemma-3-1B rises from 30.2% to 65.5%, Llama-3.2-1B from 34.5% to 38.4%, and Qwen3-0.6B falls from 83.2% to 69.8% (their Table 2; they read the Qwen3 drop as a possible collapse). So the same Llama-3.2-1B erodes little in their setup and nears saturation in ours. Their data, judge, starting rate and recipe all differ from ours. Khan et al. (2026) fine-tune four 7–9B families on medical and legal data. They write: "Base model family was likewise an unreliable predictor" (their §4.2.1). Guo et al. (2026) compare Llama-3.1-8B and Qwen2.5-7B, with no OLMo model. On Alpaca, Qwen erodes less (LoRA on 5,000 rows: 9.1% against 44.8% attack success). On Dolly with full fine-tuning, the order flips (73.6% against 66.9%). So at 7–8B, the Qwen-versus-Llama order in Guo et al. (2026) depends on the data. Fraser et al. (2025) fine-tune Llama-3.2-1B on Dolly with 4-bit LoRA for 5 epochs. Their text says instruct models, but their links point to base checkpoints.

Shallow alignment. Qi et al. (2025) argue that safety alignment sits mostly in the first few answer tokens. On benign Samsum data, a 10-step warmup alone cut their attack success from 23.4% to 3.8% (Table 9 of the ICLR version; Table 5 of arXiv v1). They credit their constrained objective mainly for the attack settings. The Scheduled recipe includes a warmup, and §5 finds that the opening switch carries about half of OLMo's erosion.

Content of benign data. Zhao et al. (2025b) fine-tune 95 role-specific LLaMA-3-8B models and find that safety falls further the more villainous the character. Vetter et al. (2026) trace the features behind emergent misalignment to text about villains and harmful agency. Fine-tuning on such human-written documents did not reliably cause it, while synthetic instruction-answer pairs made from the same content did. Narrow fine-tuning can generalise broadly (Betley et al., 2026), and answer format alone can raise erosion (He et al., 2024). Our follow-up on content (§1) will train on human-written story continuations, which sit between these results.

OLMo-2. We are not aware of prior benign-fine-tuning erosion numbers for OLMo-2 Instruct. Our check found work on refusal geometry in OLMo-2-1B, whose only fine-tuning adds refusals (Labunets, 2026), and work on OLMo-2 7B and 13B with harmful data only (Lee et al., 2026). One web search found no such numbers, and it was not exhaustive.

Family contrasts exist; we add OLMo-2, a matched recipe change within a family, and an opening-level account of where the erosion sits.

8 Conclusion

One recipe can give very different erosion across families at the same size, and a standard loss and schedule cut the erosion in one family but barely in another.

What this means in practice. First, an erosion number describes a model family, a recipe,

a dataset and a judge together: report all four, and every setting screened. Second, a fix found on one family may not carry over: the learning-rate schedule cut OLMo's erosion by about a third but barely moved Llama's, so check a fix on your own model. Third, compare families only at matched recipes, and recipes only within a family.

These measurements set up the larger study of repairing safety after benign fine-tuning. A follow-up will test whether the topic of benign data matters (dark against light fiction, at matched format and length), with its analysis plan pre-registered before any training run; we will add its results in a later version.

Limitations

- **Scope.** Every claim holds only for these models (three 1B-scale instruct families and Qwen2.5-7B), one benign dataset (Dolly), LoRA fine-tuning, these recipes, the 200 development harm prompts and these judges. No result is confirmed on held-out prompts.
- **Choices made after seeing results.** We fixed the 20–55 pp window for partial erosion after the three Qwen results, with no written reason for 20 and 55. Seed 1101 chose the setting and counts in the OLMo mean.
- **Few seeds.** The between-seed variance rests on 2 df: the sample SD's 90% range is 0.23–1.73 times the true SD. The OLMo band covers evaluation noise only. The Qwen results and the variant rows with rank 16 or half the learning rate rest on one or two seeds, and the family-by-recipe test has about 5 df.
- **What the judge measures.** U is a broad policy label that mostly tracks lost refusal (§6), and usable harm is a smaller share. HarmBench did not judge every unit, and Granite judged greedy answers only.
- **Differences between families.** The families differ in chat template, tokeniser (so the same rows give different token counts), post-training pipeline and starting over-refusal. The Qwen runs, OLMo's Constant runs and the other runs used three engine identities (Section B). We claim no cause for the family difference.
- **Recipes.** The Constant recipe is not standard practice. The token-mean recipe separates the loss weighting from the schedule, but not the schedule's warmup from its decay, which also halves the mean learning rate.
- **Human read.** One reader, who designed parts of the study, rated 50 answers. There is no second rater, and the groups are too small to compare.
- **ARC.** The ARC check is uninformative at 1B.

Ethical considerations

This paper measures how fine-tuning on benign data erodes refusal in small open models, so that practitioners can see that the model family and the training recipe change the result. It proposes no attack, and the training data is a public benign instruction dataset. We release no harmful prompt or answer text. An author read 50 model answers to harmful requests for the blind human read; no other annotators took part.

Use of AI assistants. We used AI assistants (Claude, Anthropic) extensively: to help design and pre-register the analyses, to write the analysis and run-management code, to check every reported number against the run records, and to draft and edit the text. The authors made every study decision and take full responsibility for the content.

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A Per-seed tables

Table 3 gives every seed of every run in Tables 1 and 2, the Qwen2.5-1.5B screens (Constant, rank 16 and High-dose) and Llama-3.2-1B under Constant, half LR. Tables 1 and 2 give the means.

B Settings and hashes

C Screening log

We list every setting we ran, in time order. All runs use LoRA rank 64, unless marked; §2 defines the recipes, and Section A gives every seed.

1. Qwen2.5-1.5B, High-dose, 3 seeds: mean $E = +81.8$ pp after 3 epochs. It failed the bar that answers must end (repetition loops).

2. Qwen2.5-1.5B, Constant, seed 1101: $+75.2$ pp. Seed 1102 was lost to a timer and gave no result.

3. Qwen2.5-7B, Constant, seed 1101: $+82.2$ pp.

4. Qwen2.5-1.5B, Constant, rank 16, seed 1101: $+67.8$ pp.

5. OLMo-2-1B, Constant, seed 1101: $+43.5$ pp; then seeds 1102 and 1103.

6. OLMo-2-1B, Scheduled, seed 1101: $+29.0$ pp. We pre-registered the prediction that it erodes less than the Constant recipe.

7. Llama-3.2-1B, Constant, seed 1101: $+62.5$ pp.

8. OLMo-2-1B, Scheduled, seeds 1102 and 1103; Llama-3.2-1B, Scheduled, seeds 1101–1103; OLMo-2-1B, Constant, token-mean and Scheduled, rank 16; both families under Scheduled, half LR (Table 2).

9. A Qwen2.5-1.5B Scheduled run was planned for this note, with a written prediction, and dropped before any run (9 Oct). Nothing from it exists.

10. Llama-3.2-1B, Constant, seeds 1102 and 1103 (9 Oct): $+61.3$ and $+54.0$ pp. Seed 1102 needed a second attempt: a cap gate stopped the first attempt before its job started.

11. Llama-3.2-1B, Constant, half LR, seed 1101 (9 Oct): $+48.0$ pp. Granite, judged on the pod, agrees in direction (greedy E $+47.5$ pp).

12. OLMo-2-1B, Constant, token-mean, seeds 1102 and 1103 (9–10 Oct), to complete three seeds: $+39.5$ and $+42.5$ pp.

Table 3: Every seed. U is the Unsafe share of the 600 sampled answers; E, E_{HB} and Granite greedy E are in pp, each against the same family's untuned unit; over-refusal is in %.

Recipe	Seed	U	E	E_{HB}	Over-refusal	Held-out loss	Granite greedy E
OLMo-2-1B							
untuned		3.2%			17	2.100	
Constant	1101	46.7%	+43.5	+27.7	4	1.542	+44.0
Constant	1102	46.8%	+43.7	+30.5	6	1.547	+40.5
Constant	1103	51.3%	+48.2	+32.8	4	1.545	+44.5
Constant, token-mean	1101	46.8%	+43.7	—	9	1.587	+38.0
Constant, token-mean	1102	42.7%	+39.5	—	8	1.589	—
Constant, token-mean	1103	45.7%	+42.5	—	5	1.590	—
Scheduled	1101	32.2%	+29.0	+17.8	17	1.612	+23.0
Scheduled	1102	34.2%	+31.0	+19.3	14	1.613	+25.5
Scheduled	1103	29.7%	+26.5	+14.3	18	1.613	+18.5
Scheduled, half LR	1101	23.5%	+20.3	—	18	1.634	+11.5
Scheduled, half LR	1102	24.5%	+21.3	—	22	1.635	+12.0
Scheduled, rank 16	1101	23.3%	+20.2	—	25	1.649	+9.0
Qwen2.5-1.5B							
untuned		4.7%			68	1.890	
Constant	1101	79.8%	+75.2	+46.0	3	1.459	—
Constant, rank 16	1101	72.5%	+67.8	+45.8	3	1.481	+75.5
High-dose	1101	87.3%	+82.7	—	2	1.857	+88.0
High-dose	1102	88.7%	+84.0	—	3	1.829	—
High-dose	1103	83.5%	+78.8	—	10	1.850	+85.5
Qwen2.5-7B							
untuned		4.8%			7	3.335	
Constant	1101	87.0%	+82.2	+63.5	3	1.293	+87.5
Llama-3.2-1B							
untuned		13.8%			11	2.243	
Constant	1101	76.3%	+62.5	+40.5	2	1.555	+68.5†
Constant	1102	75.2%	+61.3	—	4	1.554	+63.0†
Constant	1103	67.8%	+54.0	—	3	1.553	+57.0†
Constant, half LR	1101	61.8%	+48.0	—	5	1.566	+47.5†
Scheduled	1101	73.8%	+60.0	+35.5	7	1.619	+58.5
Scheduled	1102	70.3%	+56.5	+35.8	9	1.614	+58.5
Scheduled	1103	68.7%	+54.8	—	6	1.613	+53.0
Scheduled, half LR	1101	55.0%	+41.2	—	4	1.642	+38.0

Blank: not applicable (untuned rows). —: not recorded (HarmBench did not judge the run; Granite did not judge Qwen2.5-1.5B under Constant or High-dose seed 1102, or OLMo-2-1B Constant, token-mean seeds 1102 and 1103). High-dose rows are after 3 epochs; this recipe failed its own bar that answers must end (repetition loops), so it is a screened setting, not a clean result. Untuned HarmBench shares (pod): OLMo 3.2%, Qwen2.5-1.5B 3.2%, Qwen2.5-7B 5.3%, Llama 11.0%. Untuned Granite greedy shares: OLMo 2.5%, Qwen2.5-1.5B 3.0%, Qwen2.5-7B 4.0%, Llama 13.5% (Mac) and 13.0% (pod). †Granite on the pod, against the pod-judged untuned unit; every other Granite value is from the Mac. On the Mac, Llama's Constant seed 1101 gives +67.5.

Table 4: Training and evaluation settings. Every recipe shares the top part; the bottom part lists what each recipe changes. The release holds every recipe file with its SHA-256.

Setting	Value
Shared by every recipe	
Training code	our own training loop in PyTorch, with Hugging Face Transformers (model loading) and PEFT (LoRA)
Adapter	LoRA on the q, k, v, o, gate, up and down projections of every layer; dropout 0. Trainable parameters at rank 64: OLMo-2-1B 48.2M, Llama-3.2-1B 45.1M, Qwen2.5-1.5B 73.9M, Qwen2.5-7B 161.5M
Optimiser	AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, no weight decay; gradient-norm clipping at 1.0
Batch	128 rows per optimiser step (micro-batches of 4 rows with 32 accumulation steps; Qwen2.5-7B: 2 rows and 64 steps); one epoch of 4,800 rows is 38 steps
Loss	next-token negative log-likelihood on the answer and its end-of-turn token only; the prompt is masked
Precision and length	base weights in bfloat16; adapters and the loss in float32; SDPA attention; at most 3,072 tokens per training row (1,536 in the prompt)
Harm evaluation	200 development harm prompts; 3 sampled answers each (temperature 0.7, top-p 1.0) and one greedy answer; at most 2,048 new tokens
Other measures	over-refusal on 100 benign prompts (greedy answers); held-out loss on 200 Dolly rows; ARC forced choice on 189 questions
Judges	Qwen3Guard-Gen-4B on every answer (the main judge); the HarmBench Mistral-7B classifier on sampled answers; Granite Guardian 3.3 8B on greedy answers
What each recipe changes	
Constant	learning rate 10^{-4}, constant; per-row loss mean; rank 64, alpha 128
Constant, token-mean	as Constant, with a token-mean loss
Constant, half LR	as Constant, with learning rate 5×10^{-5}
Constant, rank 16	as Constant, with rank 16 and alpha 32
Scheduled	peak learning rate 10^{-4}: a 4-step linear warmup, then cosine decay to 0 at the end of epoch 1; token-mean loss; rank 64, alpha 128
Scheduled, half LR	as Scheduled, with peak 5×10^{-5}
Scheduled, rank 16	as Scheduled, with rank 16 and alpha 32
High-dose (a Qwen2.5-1.5B screen)	as Constant, with learning rate 2×10^{-4}, 8 rows per step (600 steps per epoch) and 3 epochs

Table 5: Revisions and SHA-256 hashes. Model and judge revisions are Hugging Face commit hashes. Table 4 gives the training and evaluation settings.

Item	Revision or SHA-256
Models	
OLMo-2-0425-1B-Instruct	`48d788eca847d4d7548f375ad03d3c9312f6139e`
Qwen2.5-1.5B-Instruct	`989aa7980e4cf806f80c7fef2b1adb7bc71aa306`
Qwen2.5-7B-Instruct	`a09a35458c702b33eeacc393d103063234e8bc28`; tokenizer from the Qwen2.5-1.5B-Instruct revision above (identical files)
Llama-3.2-1B-Instruct	`9213176726f574b556790deb65791e0c5aa438b6`; tokenizer and chat template from Llama-3.1-8B-Instruct at `0e9e39f249a16976918f6564b8830bc894c89659`
Judges	
Qwen3Guard-Gen-4B	`6ec42827da0c1ff11e7a49dc269d2e810d27e108`
HarmBench-Mistral-7b-val-cls	`51182c7cdaf9b2e5f05b745b81b047b67e7384a1`
Granite Guardian 3.3 8B, Q4_K_M	granite-guardian-3.3-8b-GGUF at `8c87479a2dc329e4e362a6e5dda7e7de3e0372b2`; file SHA-256 `b4fa861b5163373413d9659c22e344e6206baad824ead17226b2058e54d14587`
Prepared Dolly data (SHA-256 of each manifest)	
Qwen2.5 (1.5B and 7B)	`9247c4bbccf212b02be7f221e2ee451d17ebf643a39a44db6e3bed6fb4d750f2`
OLMo-2-1B	`9ed61826f1634dcab5a7385f18556b108b3ba8db95fc17215fa4cf4f6cb993b9`
Llama-3.2-1B	`c52a50afddcfcfa818512ac7f1536c34c959774e815095bd30f69320b083d55a`
Engine identities (code SHA-256)	
Qwen2.5 runs	`c52c0ebc83793fc552f84908e83542d5e46543459fcafb7be2fc9c7ff42ac597`
OLMo-2-1B, Constant	`552538748acec1a8ffd1ee976b9eb3c3fdfe9ca01b8148ee8810291409357441`
Llama-3.2-1B runs, and the other OLMo-2-1B runs	`e49a978c5fa27c3ba4bb1570ac8b24501e09bf8db5c35e31ffba90bf775cdc06`
Analysis	
Opening word list, frozen before any result (§5)	`bbbc71e9eebb62876430ffb5e0f54113c3818984b8369924b7bd2212c4c1ca17`
Reading bars, with the 20–55 pp window (§3)	`c0b6f2aae43285de632a175adb1e913cd31f37bc6ac25fb56e4468ff3a1208ea`
Logit-scale prediction for the Scheduled recipe on Qwen2.5-1.5B, written before the run was dropped (§6)	commit `c868608a08ea1192e1e8bf489488631a4903beaa`, 2026-10-08 17:14 UTC